

# Detecting Suspicious Internal Activity Using After Hours Logon Patterns

## 1. Problem Statement

This project focuses on building a simple and effective machine learning model to detect potential internal threats within an organization based on employee logon behavior. As organizations increasingly rely on digital infrastructure, the risk of internal threats has become a critical cybersecurity concern. Specifically, the model aims to identify suspicious activity by flagging employees who log into their machines during after hours. The goal is to simulate a real-world security monitoring tool that could assist a CISO (Chief Information Security Officer) team. I chose this dataset because it was ideal for this project and good practice for my CISO internship at Bloomberg this summer!

## 2. Dataset Description

The project utilizes the logon.csv file from the CERT Insider Threat dataset (provided by Carnegie Mellon University). This file contains:

- **id:** Event identifier
- **date:** Timestamp of the logon or logoff event
- **user:** User ID
- **pc:** Computer ID
- **activity:** Type of event (Logon or Logoff)

This dataset was ideal because it's structured, small in size, and directly relevant to internal monitoring systems. It also includes enough data to train models effectively while avoiding extensive preprocessing.

|   | user         | day        | after_hours_logon_flag | usb_used_flag | urls_after_hours | is_suspicious |
|---|--------------|------------|------------------------|---------------|------------------|---------------|
| 0 | DTAA/AAA0371 | 2010-01-04 | False                  | False         | 0.0              | 0             |
| 1 | DTAA/AAA0371 | 2010-01-05 | False                  | False         | 0.0              | 0             |
| 2 | DTAA/AAA0371 | 2010-01-06 | False                  | False         | 0.0              | 0             |
| 3 | DTAA/AAA0371 | 2010-01-07 | False                  | False         | 0.0              | 0             |
| 4 | DTAA/AAA0371 | 2010-01-08 | False                  | False         | 0.0              | 0             |

### 3. Data Preprocessing & Feature Engineering

Steps taken:

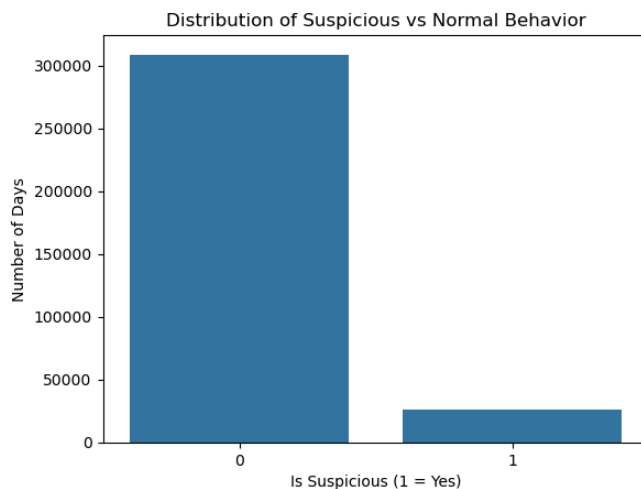
- Converted the date column to a datetime object using `pd.to_datetime()`
- Created a new boolean feature: `after_hours_logon_flag` based on whether a user logged in before 7 am or after 6 pm
- Grouped the data by user and day to summarize each user's behavior
- Created the target label `is_suspicious` for if a user logged in after hours on a given day, that day was labeled as suspicious (1), otherwise normal (0)

This resulted in a binary classification problem using a single engineered feature.

### 4. Exploratory Data Analysis (EDA)

EDA was conducted using Seaborn and Matplotlib:

- **Class Distribution Plot:** A count plot showed that most user-days were not flagged as suspicious. This helped illustrate the class imbalance that's typical in internal threat detection.

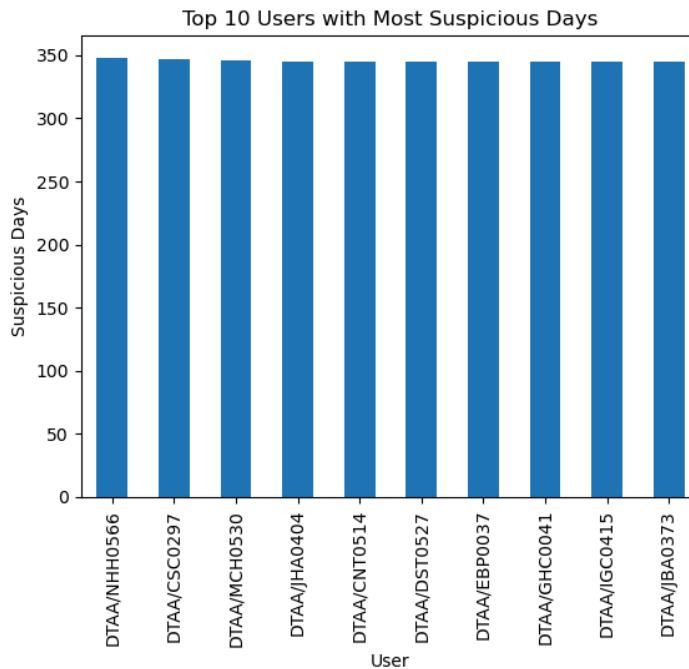


- **Summary Statistics:** Using `describe()`, we found that approximately 7.8% of user-day pairs were labeled as suspicious, which matches expectations for rare internal anomalies.

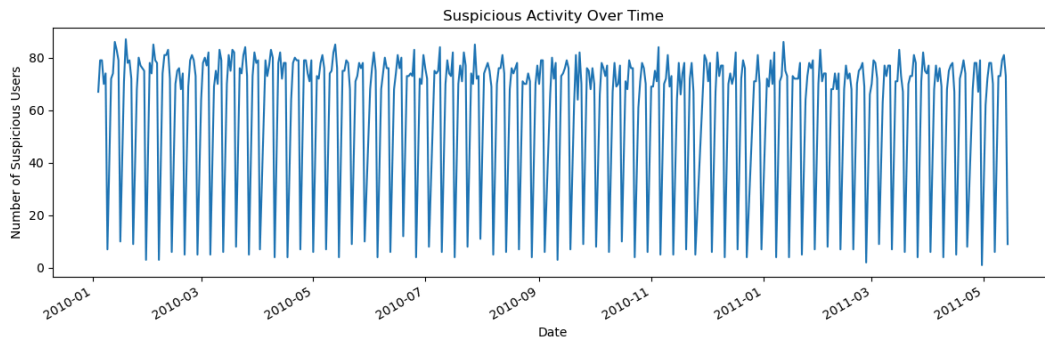
Summary stats:

|       | day                           | is_suspicious |
|-------|-------------------------------|---------------|
| count | 334905                        | 334905.000000 |
| mean  | 2010-09-04 05:31:08.607516160 | 0.078291      |
| min   | 2010-01-04 00:00:00           | 0.000000      |
| 25%   | 2010-05-03 00:00:00           | 0.000000      |
| 50%   | 2010-09-01 00:00:00           | 0.000000      |
| 75%   | 2011-01-10 00:00:00           | 0.000000      |
| max   | 2011-05-14 00:00:00           | 1.000000      |
| std   | NaN                           | 0.268629      |

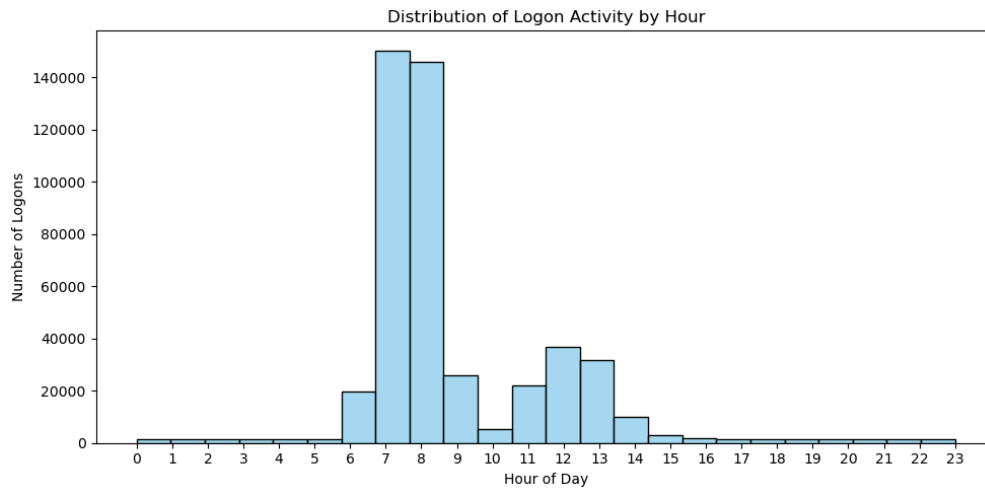
- **Top 10 Users Plot:** A bar chart of the top 10 users with the most suspicious logins highlighted specific individuals whose after-hours behavior was most frequent. This provided insight into which users stood out as behavioral outliers.



- **Suspicious Activity Over Time:** A line plot tracked the number of suspicious events per day across the entire dataset. This revealed temporal trends in behavior, such as spikes on certain days, suggesting either targeted activity or systemic changes.



- **Logon Hour Distribution:** A histogram showing logon events by hour of the day confirmed that most logons occurred between standard business hours (7 AM – 6 PM), and visually validated that our after-hours flag was meaningful.



These visuals collectively supported the decision to use after-hours logon activity as a strong predictor.

## 5. Model Selection and Implementation

Three models were selected for simplicity and interpretability:

### a. Logistic Regression

Serves as a baseline linear classifier. Achieved perfect results with the single feature.

### b. Random Forest Classifier

A robust ensemble method that performs well with minimal tuning. Also achieved perfect performance.

### c. Decision Tree Classifier

Useful for explaining model decisions. Matched the performance of the other models.

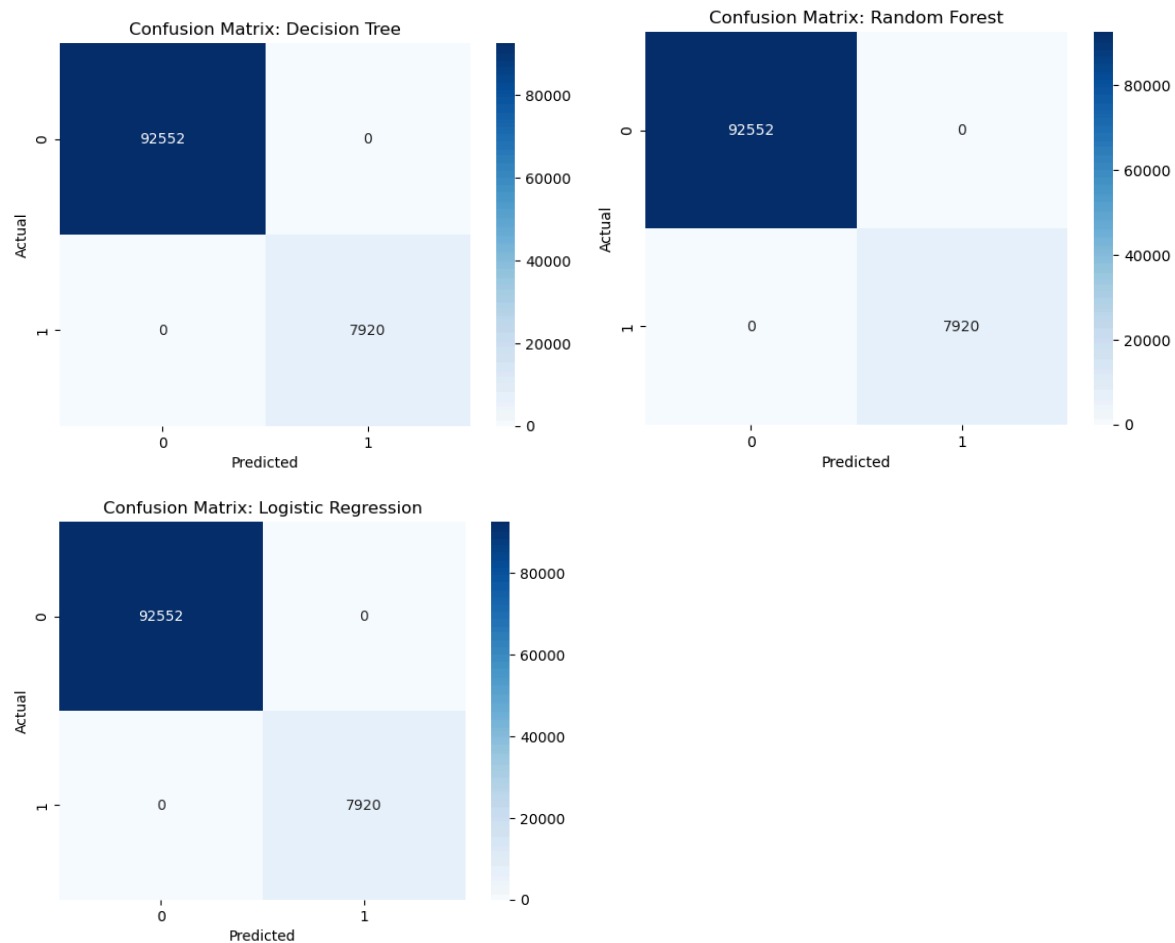
All models were trained using scikit-learn and evaluated using classification reports and confusion matrices.

## 6. Model Evaluation

All models produced perfect classification results:

- Accuracy: 100%
- Precision, Recall, F1-Score: 1.00 for both classes
- Confusion matrices showed zero false positives and zero false negatives

While these results seem ideal, it's important to note the simplicity of the dataset and task. Real-world data would likely yield more noise and lower metrics.



## 7. Strengths and Limitations

### Strengths:

- Clear and well-defined classification problem
- Fast runtime and low complexity
- Perfect classification using interpretable models

### Limitations:

- The model used only one engineered feature
- Synthetic dataset with clear patterns, not realistic in messy environments
- Lacked complexity like time-series modeling or anomaly scoring

Future improvements could include integrating file access, USB usage, and web traffic logs.

## 8. Conclusion

This project demonstrated how machine learning can be used to detect potential internal threats based on a single, interpretable feature: after-hours logon activity. Using a small, clean dataset and simple modeling techniques, I was able to build a system that effectively flagged suspicious behavior. All three models: Logistic Regression, Random Forest, and Decision Tree, performed with perfect accuracy, precision, and recall. This outcome wasn't surprising given how clearly the feature separated the two classes, but it still showed how one variable can go a long way.

While the results were strong, they also point to a bigger takeaway: detecting insider threats depends heavily on the quality and relevance of the features you use. In this case, after-hours logins worked really well as a signal, but in real environments, behavior is usually more nuanced and harder to label. Still, this project gave me a useful starting point for understanding how even minimal inputs can produce meaningful results when approached thoughtfully.

The project also ties closely to the kind of work done by CISO teams and security analysts, where unusual patterns in user access behavior are often the first signs of risk. So, beyond just meeting the DAT 412 requirements, this felt like valuable prep for my upcoming internship at Bloomberg. If I were to continue this project, I'd expand it to include other types of behavior like web browsing, file access, or USB activity, and explore more advanced approaches like anomaly detection or time-based modeling to make it more realistic.

## 9. References

- CERT Insider Threat Dataset:  
<https://www.cmu.edu/engage/insider-threat/tools/data-sets.html>  
Lindauer, Brian (2020). Insider Threat Test Dataset. Carnegie Mellon University. Dataset. <https://doi.org/10.1184/R1/12841247.v1>